MICRO COMPUTER INSTITUTE

Python Programming

Foundation Study Pack - Volume 1

PYTHON | Duration: 4 Months

Learn - Practise - Apply
A structured bilingual-support handbook for classroom, lab and online learners.

Student use: Read each module, complete the lab activity, save evidence in your portfolio, and discuss doubts with your instructor.

Hindi support (Roman): Har module ko padhen, practical lab activity poori karen, apna work save karen aur doubt teacher se poochhen.

Course Roadmap

This 4 Months programme is organised into 8 learning modules. The sequence may be adjusted by the instructor to match the batch calendar.

No.	Module	Learning focus
1	Programming Basics	Algorithms, variables, data types, input and output.
2	Decision Making	Boolean logic, if, elif and nested conditions.
3	Loops	for, while, range and loop-control statements.
4	Collections	Strings, lists, tuples, sets and dictionaries.
5	Functions	Parameters, return values, scope and reusable modules.
6	Files & Errors	Text/CSV files, exceptions and validation.
7	Object-Oriented Basics	Classes, objects, attributes and methods.
8	Final Project	Create a tested command-line student or inventory application.

Module 1: Programming Basics

Algorithms, variables, data types, input and output.

Key ideas

• Explain the purpose and essential vocabulary of programming basics.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates programming basics. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 2: Decision Making

Boolean logic, if, elif and nested conditions.

Key ideas

• Explain the purpose and essential vocabulary of decision making.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates decision making. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 3: Loops

for, while, range and loop-control statements.

Key ideas

• Explain the purpose and essential vocabulary of loops.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates loops. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 4: Collections

Strings, lists, tuples, sets and dictionaries.

Key ideas

• Explain the purpose and essential vocabulary of collections.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates collections. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 5: Functions

Parameters, return values, scope and reusable modules.

Key ideas

• Explain the purpose and essential vocabulary of functions.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates functions. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 6: Files & Errors

Text/CSV files, exceptions and validation.

Key ideas

• Explain the purpose and essential vocabulary of files & errors.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates files & errors. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 7: Object-Oriented Basics

Classes, objects, attributes and methods.

Key ideas

• Explain the purpose and essential vocabulary of object-oriented basics.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates object-oriented basics. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 8: Final Project

Create a tested command-line student or inventory application.

Key ideas

• Explain the purpose and essential vocabulary of final project.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates final project. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Practical Workbook

Complete these tasks in order. The instructor may adapt names and sample data, but every learner should produce original work.

Practical 1. Create and save a correctly named practice file for Programming Basics. Include a short note describing what you learned.

Practical 2. Create and save a correctly named practice file for Decision Making. Include a short note describing what you learned.

Practical 3. Create and save a correctly named practice file for Loops. Include a short note describing what you learned.

Practical 4. Create and save a correctly named practice file for Collections. Include a short note describing what you learned.

Practical 5. Create and save a correctly named practice file for Functions. Include a short note describing what you learned.

Practical 6. Create and save a correctly named practice file for Files & Errors. Include a short note describing what you learned.

Portfolio checklist

- Use folder format: CourseCode / StudentName / Module / Activity.
- Keep editable source files plus exported PDF or screenshot.
- Do not copy another student work; mention sources used.
- Back up important work in two locations.

Assignments & Assessment

Assignment 1: Concept Check

Write 300-500 words explaining three important concepts from Programming Basics and Decision Making. Add one real-life example.

Assignment 2: Practical Submission

Complete a combined task using Loops and Collections. Submit source file, output and a one-page process note.

Assignment 3: Problem Solving

Identify a realistic institute or small-business problem that can be improved using Python Programming. Propose steps, tools, risks and success measures.

Assignment 4: Final Project

Build a complete project using at least four course modules. Test it, prepare screenshots, and present the result in 5-7 minutes.

Criterion	Weight
Accuracy and completeness	30%
Practical application	30%
Presentation and file organisation	20%
Originality, explanation and safe practice	20%

Revision & Safe Learning Guide

- Review class notes within 24 hours and practise again without watching the demonstration.
- Use legal software and trusted downloads. Scan external drives and downloaded files.
- Never share passwords, OTPs, private documents or student data.
- Verify AI-generated or internet information before using it in assignments.
- Use clear filenames, version numbers and regular backups.
- Ask for help when an error could damage data, equipment or accounts.

Quick viva questions

1. What is the purpose of Programming Basics, and what common mistake should a beginner avoid?
2. What is the purpose of Decision Making, and what common mistake should a beginner avoid?
3. What is the purpose of Loops, and what common mistake should a beginner avoid?
4. What is the purpose of Collections, and what common mistake should a beginner avoid?
5. What is the purpose of Functions, and what common mistake should a beginner avoid?
6. What is the purpose of Files & Errors, and what common mistake should a beginner avoid?

Support

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